Pre

Bachelor in Computer Applications
Course Title: Data Structures and Algorithms
Code No: CACS 201
Semester: III

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. **[6×5 = 30]**

2. Define data structure. How linear data structure is different from non-linear data structure? Explain.
3. List any two applications of stack. Convert A* B-(C*D-A)/(C-D)+E into postfix notations using stack.
4. Define priority queue. How circular queue is different from double ended queue? Explain.
5. Use the heap sort algorithm to sort the following data items: 12, 1, 17, 31, 24, 27, 19, 15, 18
6. What is hashing? How collision resolution can be done using separate chaining method? Explain.
7. How quick sort works? Write a function to perform quick sort.
8. Differentiate between internal and external sorting? What do you mean by divide and conquer algorithm? Explain with example.

Group C

Attempt any TWO questions. **[2×10 = 20]**

9. Explain different types of graph. Explain Kruskal's algorithm in detail.
10. Explain balanced binary tree and AVL Tree with proper examples. Create an AVL tree from the following: 4, 6, 12, 19, 15, 25, 13, 8, 3, 7, 11, 17, 31.
11. How binary search is different from sequential search? Explain. Consider a hash table of size 9 and insert the keys 9, 19, 32, 10, 73, 85, 50 using linear probing.

Tribhuvan University
Faculty of Humanities & Social Sciences
Patan Multiple Campus
Pre-board Examination

Bachelor in Computer Applications
Course Title: System Analysis and Design
Code No: CACS203
Semester: III

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Pass Marks: 24
Time: 3 hours

SET A

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions. [6×5 = 30]

1. Define system? What are its basic characteristics and components.
2. Explain outsourcing with its advantages and disadvantages.
3. What are the different factors influencing maintenance cost
4. Define case tools. Explain different types of case tools
5. SDLC is analogous to maintenance process. Justify
6. What are the different types of installation. Explain
7. What is JAD? Explain participants of JAD.
8. Define disruptive technology. Explain any three disruptive technology that has changed your life style.

Group "C"

Attempt any TWO questions. [2×10 = 20]

9. Suppose you are project manager for patient's information system for BIR hospital. Which model would you choose Waterfall model or Agile? Justify your choice (7+3)
10. What are the rules to draw DFD. Draw DFD for movie ticket booking system. (2+8)
11. What are different techniques to access economic feasibility. 10

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SET B

Group B

Answer any 6 questions 6×5=30

11 Define system. Explain different types of system.
12 Define off-the shelves software. Explain different origins of software
13 Define corporate strategic planning. Explain how system development helps in achieving the goal of business house
14 Define steering committee. What are the different types of evaluation criteria.
15 Explain different types of feasibility assessment to be taken.
16 Define BPP. Explain the components of BPP with format
17 Define data modeling. Explain conceptual data modeling with example
18 Define normalization. Explain normalization upto 3NF with example

Group C

Answer any 2 questions 2×10=20

1 Define Context diagram. List out different types of symbols used DFD. Draw DFD for online hotel room booking system. (1+2+7)
2 Define user interface. Explain different types of interfaces that can be implemented in modern system. Explain different types of menus. (2+3+5)
3 What are the phases that is incorporated within implementation. Explain different types of system testing. (3+7)

Group-B

Attempt any SIX questions **[5X6=30]**

1. What are java Buzzwords? Write a java program to find simple interest. Use command line argument to take input.
2. What is recursive function? Write a java program using recursive and iterative method to find n^{th} Fibonacci number.
3. What is jagged array? Given an array, write a java program to right rotate it by k elements. Let elements be 1,2,3,4,5,6,7,8,9,10. If k=3, after right rotation it becomes 8,9,10,1,2,3,4,5,6,7.
4. What is final keyword? How will you implement multiple inheritance using interface? Explain with program.
5. Write a simple java program that reads data from one file and writes data to another file using Character stream class.
6. What is autoboxing and unboxing in java? Explain difference between HashMap and HashTable.?
7. What is menu? Write a program to create a menu named "File" with menu items "New," "Save," and "Exit".

Group-C

Attempt any TWO questions **[10X2=20]**

1. What are keywords used in Exception handling in java? Write java program that will read balance and withdrawal amount from keyboard and display remaining balance on screen if balance is greater than withdraw otherwise throw an Application Exception with appropriate message.

   model view control
2. What is MVC design pattern? Write a java program using swing component to find largest and smallest number among three number. Use text fields for inputs and output. Your program should display the result when user presses a button.
3. What is difference between executeUpdate() and executeQuery() method ? Write a java program to connect database **Bank** and insert 5 customer record (Account_no, Name, Address, Balance) in **Customer** table and Display the customer record whose balance is greater than 5000.

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...urse Title: Web Technology
Code No: CACS 205
Semester: 3rd

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any six questions. **[6*5 = 30]**

2. Discuss HTML table tags and its attribute with its example
3. What is XML? Write rules to create XML document.
4. Why we need XML schema? Explain different type of element into XML Schema.
5. What is CSS? Discuss different types of CSS with example.
6. What is Cookie? Explain working mechanism of session
7. Explain multitier technology with its advantage and disadvantage.
8. What is XPath? Write XSLT code to display your books information (title, price, ISBN, subject code) stored into XML.

Group C

Attempt any two questions. **[2×10 = 20]**

9. Write server-side script to create, validate and store into database details of student of loksewa aayog form with following information.
   [5+5]
   a. Name
   b. Citizenship no
   c. Email
   d. Applied Position Plus Two Faculty (Humanities, Science, Law, Management, Education.
10. Compare XML Schema with DTD. Write XML schema to describe student as root element, name as child element with first_name,middle_name,last_name, city as child element with its content "KTM","PKR","LPT" and rollno as its attribute.
11. Create web page with following details
    a. Create basic page structure
    b. Add header section to the top of page which contain logo and search bar
    c. Create two column layouts of (25%,75%) width, and put unordered list into first one and image gallery into second one
    d. Add footer section, which includes address of company and copyright information.
    e. Also add some basic CSS using external stylesheet

Group-B

Pre

Attempt SIX the questions

5x6=30

2. Write the application of statistics in computer science. Describe the scope and limitations of the statistics.

3. For the data given below and use it, to find the value of mean, median and mode.

Daily wages (Rs.) (X)	20 - 40	40 - 60	60 – 80	80 - 100	100 - 120
No. of workers (f)	41	51	64	38	7

4. Define correlation coefficient? Calculate the Karl-Pearson's coefficients of correlation between price and supply of a commodity from the data given below:

Price (Rs.)	17	18	19	20	21	22	23	24	25	26
Supply (Rs.)	38	37	38	33	32	33	34	29	26	23

5. Define regression. The technician now varies the temperature (^{0}C) while keeping other conditions as constant as possible and obtain the following results:

Yield (Y)	125	128	130	132	134
Temperature (X)	65	75	85	85	90

(i) Using least square method and construct the regression line of yield on temperature (ii) Predict the yield when temperature is 95. (iii) Find the coefficient of determination.

6. Define Poisson distribution. Fit a Poisson distribution to the following data:

Defects (X = x)	0	1	2	3	4	5
Number of pages	142	156	69	27	5	1

$\mu = \lambda$

$\sigma =$

7. In measuring reactions time, a psychologist estimates that the standard deviation is 0.15 seconds. How large a sample of measurement must be taken in order to be 95% confident that the error of his estimate will nor exceed 0.012 seconds.

$Z_{\alpha/2} = Z_{0.05/2} = 1.96.$

$d =$

8. Define sampling. A population consists of the four numbers 1, 3, 5, 9. (i) Write down all possible sample size of two without replacement. (ii) Show that sample mean is an unbiased estimate of population mean.

$n_0 = \left(\frac{Z_{\alpha/2} \cdot \sigma}{d}\right)^2.$ 0.95 : d

=

Group-C

10x2=20

Attempt SIX the questions

9. What do you mean by statistics? J.J. Thomson discovered the electron by isolating negatively charged particles for which he could measure the mass-charge ratio. This ratio appeared to be constant over a wide range of experimental conditions and, consequently, could be characteristics of new particles. His observation, from two different Cathode ray tubes that used air as the gas.

Tube I	0.57	0.34	0.43	0.32	0.48	0.40	0.40
Tube II	0.53	0.47	0.47	0.51	0.63	0.61	0.48

Which tube is more homogeneous?

10. Define standard normal distribution. The mean yield for one-acre plot is 662 kilos with st.dev. 32 kilos. Assuming normal distribution, how many one-acre plots in a batch of 1000 plots would you expect to have yield (i) over 700 kilo , (ii) below 650 kilos and, and, (iii) between 550kilos and 750 kilos.

11. What do you mean by ANOVA? Given the data below; test the hypothesis that the mean of three samples are equal.

Sample-1	40	50	60	
Sample-2	70	65		
Sample-3	45	38	60	42

BEST OF LUCK!!!